

# Embed a shape-memory alloy lattice within the wall of each catheter lumen that self-expands to a preset diameter after insertion, creating an internal scaffold that resists lumen narrowing while keeping the catheter's external profile unchanged.

## Executive Technology Brief

**Domain:** fluidics\_hydraulic

**Technology Maturity:** ENGINEERING\_DEFINITION

**Transfer Posture:** Sponsored Validation

**Problem:** Documented problem 'lumen occlusion' concerning multi-lumen drainage catheter. User-submitted engineering problem (session input): "Why do multi-lumen drainage catheters lose flow, and would a narrower lumen resist occlusion? (baseline comparison)". [fda\_maude] 100 records retrieved live (1919-2076). Illustrative records: 'MAUDE report 100000: Injury [VITAL SIGNS]'; 'MAUDE report 10000010: Injury [RESERVOIR 3ML MMT-332A]'. LIMITS STAMPED INLINE: voluntary/regulatory reports; counts are NOT incidence rates; causality UNVERIFIED (Art. XXI.5). [europepmc] 10 records retrieved live (2023-2026). Illustrative records: 'Is Fibrin a Suitable Matrix for Small-Diameter Bioartificial Vascular Grafts? An In Vitro Short-Term Hemocompatibility Study.'; 'Urinary catheters: state of the art and future perspectives - a narrative review.'. LIMITS STAMPED INLINE: literature context establishes the problem class, not event rates. Problem-existence basis: live records retrieved.

**Mechanism:** Self-expanding metal lattice (as used in aerospace deployable structures) exerts a constant outward radial force that preserves conduit diameter under external pressure or internal blockage.

**Architecture:** PROPOSED architecture (no physical system exists): flow path / lumen architecture; pressure regulation element; sensing / feedback element (if closed-loop); termination interfaces (patient / reservoir) — realizing: Embed a shape-memory alloy lattice within the wall of each catheter lumen that self-expands to a preset diameter after insertion, creating an internal scaffold that resists lumen narrowing while keeping the catheter's external profile unchanged.

**Governing Model:** Fluid mechanics / hydraulics governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

## Key Equations:

FLUID-001: Hagen-Poiseuille (laminar pipe flow). Expression:  $Q = (\pi * r^4 * dP) / (8 * \mu * L)$ .

Applicability to this invention: APPLICABLE; holds under: Newtonian fluid, laminar regime, rigid circular lumen, fully developed flow. Assumptions: CSF approximated as Newtonian (valid at shunt flow rates); no wall compliance effects. Engaged variables:  $r$ ,  $dP$ ,  $L$ .

FLUID-002: Reynolds number. Expression:  $Re = (\rho * v * D) / \mu$ . Applicability to this invention:

APPLICABLE; holds under: circular section; single-phase flow. Assumptions: transition criterion  $Re \sim 2300$  is a MODEL\_DERIVED engineering rule of thumb. Engaged variables:  $D$ .

FLUID-003: Orifice flow (sharp-edged restriction). Expression:  $Q = C_d * A * \sqrt{2 * dP / \rho}$ .

Applicability to this invention: APPLICABLE; holds under: turbulent orifice flow;  $C_d$  treated as MODEL\_DERIVED until measured. Assumptions:  $C_d \sim 0.6-0.65$  is an engineering prior, not a measurement. Engaged variables:  $dP$ .

**Kill Condition:** killer experiment fails: falsification\_test\_from\_candidate (decision scores: EIG 0.4557178723356231, EIG-per-cost 0.4557178723356231; the falsification test carries the pre-registered pass/fail before any transfer)

Current State: ENGINEERING\_DEFINITION. No physical prototype. No clinical validation.

*Disclosure: This dossier distinguishes established evidence from proposals and unresolved questions.*